

Tirukandyur Sowmith

✉ sowmithtk@gmail.com

☎ +91-8309115971

🌐 linkedin.com/in/sowmith-tirukandyur

🐙 github.com/TIRUKANDYURSOWMITH



PROFESSIONAL SUMMARY

Detail-oriented AI Data Science M.Tech student specializing in building predictive models and data-driven systems. Strong foundation in Python, Deep Learning, and NLP, complemented by industry experience at Intel. Quick learner actively expanding into LLMs and advanced Computer Vision to solve complex, real-world problems.

EDUCATION

IIITDM KANCHEEPURAM

M.Tech in CSE Artificial Intelligence and Data Science

2025 – 2027

CGPA: 8.75 / 10.0

Malla Reddy College of Engineering and Technology (MRCET)

B.Tech in Artificial Intelligence and Data Science

2021 – 2025

CGPA: 8.5 / 10.0

INTERNSHIP EXPERIENCE

Machine Learning Intern - Intel (Remote)

June 2023

- Developed a high-accuracy Natural Language Processing (NLP) text classification model to detect deceptive text patterns, achieving a 95% peak accuracy.
- Engineered the underlying data pipeline to handle large-scale datasets, applying advanced text preprocessing, cleaning, and TF-IDF vectorization using Scikit-learn and Pandas.
- Validated model robustness and classification effectiveness by implementing comprehensive evaluation metrics, including confusion matrices and precision-recall analysis.

PROJECTS

LLM-Driven Defect Knowledge Base & Recommendation Engine

Jan 2026

- **Data Processing & Storage:** Engineered a vector-based semantic search pipeline by generating embeddings from unstructured technical fault logs and storing them in a vector database (e.g., **ChromaDB/FAISS**).
- **AI Integration:** Utilized **LangChain** to build a **Retrieval-Augmented Generation (RAG)** framework, connecting the technical defect database to an open-source Large Language Model.
- **Diagnostic Output:** Automated the mapping of input defect symptoms to historical probable causes, generating concise, actionable repair recommendations and diagnostic summaries.

Plant Disease Vision System: Classical ML to Transformers

Nov 2025

- **Baseline ML Engineering:** Extracted manual image features using **GLCM** and **LBP**, applying **PCA** for dimensionality reduction to train baseline classification models including **SVM**, **KNN**, and **Random Forest**.
- **Deep Learning Optimization:** Advanced to deep learning by optimizing a 10,000-image dataset via **OpenCV** pipelines and fine-tuning a **DenseNet201** CNN, significantly improving validation accuracy.
- **Transformer Implementation:** Culminated the project by designing and training a **Vision Transformer (ViT)**, successfully proving the superior spatial pattern recognition of transformer architectures over traditional CNNs.

Predictive Analytics for Cardiovascular Risk Assessment

Jul 2024

- **Model Development:** Built an end-to-end classification system leveraging a **Random Forest** algorithm to predict disease risk based on a diverse set of patient health parameters.
- **Data Infrastructure:** Designed a local data storage and retrieval workflow utilizing **SQLite3** to efficiently manage structured health records.
- **Feature Analysis:** Conducted comprehensive exploratory data analysis (EDA) using **Seaborn** and **Matplotlib** to visualize feature correlations and interpret the primary factors driving the model's predictions.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C++, Java, R
- **Frameworks/Tools:** PyTorch, TensorFlow, LangChain, Hugging Face, OpenCV, Scikit-learn, Git
- **Databases & Infrastructure:** MySQL, ChromaDB (Vector DB), SQLite3
- **Concepts:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP, Computer Vision, Semantic Search, Traditional ML